Walmart : Confidence Interval and CLT

Ritesh Charan Raj ND

```
# Importing the necessary libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm
```

```
# converting data into dataf

walmart = pd.read_csv("walmart_data.csv")
```

```
# making an copy of the dataset

df = walmart.copy()
```

```
# Top 5 rows of the dataframe

df.head()
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Pur
0	1000001	P00069042	F	0-17	10	A	2	0	3	
1	1000001	P00248942	F	0-17	10	A	2	0	1	
2	1000001	P00087842	F	0-17	10	A	2	0	12	

```
# No of rows & columns

df.shape
```

```
(550068, 10)
```

```
# Data info

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   User_ID                     550068 non-null  int64
 1   Product_ID                  550068 non-null  object
 2   Gender                      550068 non-null  object
 3   Age                         550068 non-null  object
 4   Occupation                  550068 non-null  int64
 5   City_Category               550068 non-null  object
 6   Stay_In_Current_City_Years  550068 non-null  object
 7   Marital_Status              550068 non-null  int64
 8   Product_Category            550068 non-null  int64
 9   Purchase                    550068 non-null  int64
dtypes: int64(5), object(5)
memory usage: 42.0+ MB
```

```
# Checking of null values

df.isna().sum()
```

```
User_ID                       0
Product_ID                    0
Gender                        0
Age                           0
Occupation                    0
City_Category                 0
Stay_In_Current_City_Years    0
Marital_Status                0
Product_Category              0
Purchase                      0
dtype: int64
```

```
# Duplicate values check

df.duplicated().sum()
```

```
0
```

```
# Uniques values of each columns

df.nunique()
```

```
User_ID                        5891
Product_ID                     3631
Gender                            2
Age                               7
Occupation                       21
City_Category                     3
Stay_In_Current_City_Years        5
Marital_Status                    2
Product_Category                 20
Purchase                      18105
dtype: int64
```

```
# Convert all columns (except Purchase) to categorical type in the DataFrame

for _ in df.columns[:-1]:
 df[_] = df[_].astype('category')

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   User_ID                     550068 non-null  category
 1   Product_ID                  550068 non-null  category
 2   Gender                      550068 non-null  category
 3   Age                         550068 non-null  category
 4   Occupation                  550068 non-null  category
 5   City_Category               550068 non-null  category
 6   Stay_In_Current_City_Years  550068 non-null  category
 7   Marital_Status              550068 non-null  category
 8   Product_Category            550068 non-null  category
 9   Purchase                    550068 non-null  int64
dtypes: category(9), int64(1)
memory usage: 10.3 MB
```

```
df.describe()
```

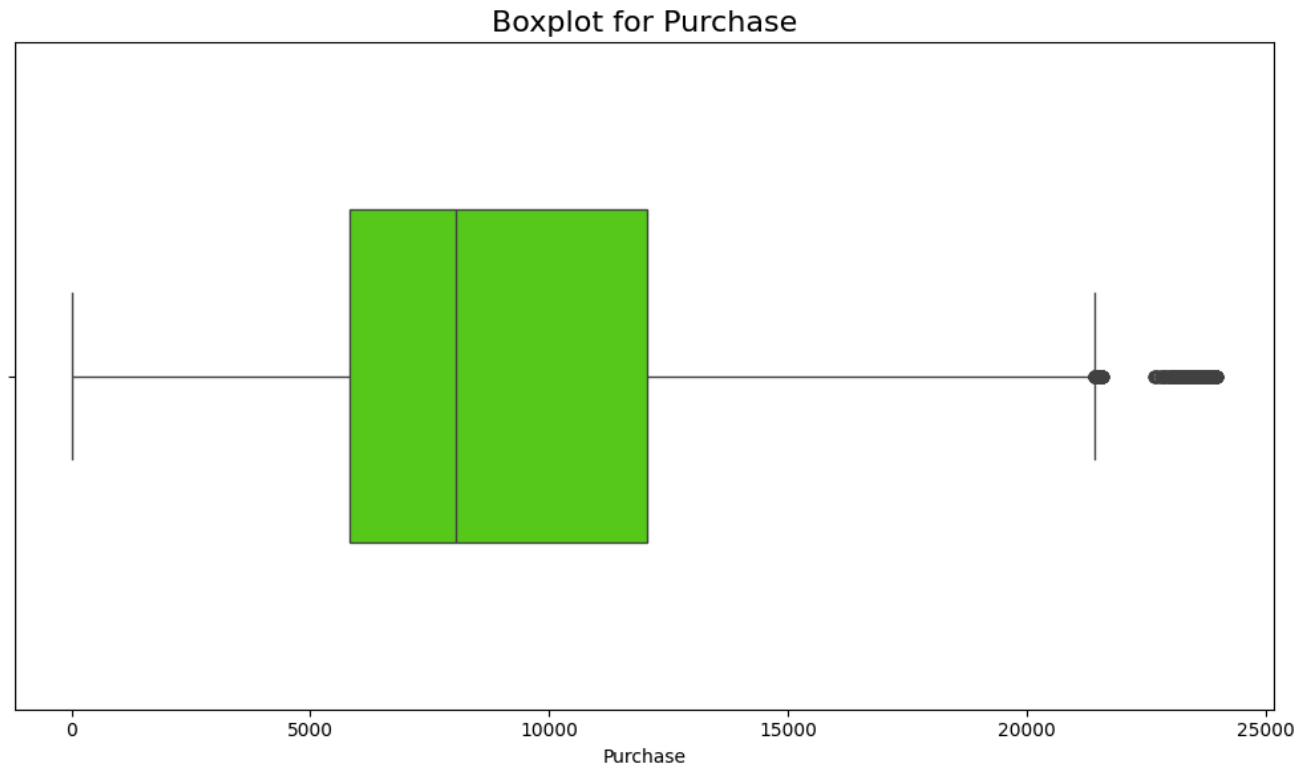
	Purchase
count	550068.000000
mean	9263.968713
std	5023.065394
min	12.000000
25%	5823.000000
50%	8047.000000
75%	12054.000000
max	23961.000000

```
df.describe( include = 'category').T
```

	count	unique	top	freq
User_ID	550068	5891	1001680	1026
Product_ID	550068	3631	P00265242	1880
Gender	550068	2	M	414259
Age	550068	7	26-35	219587
Occupation	550068	21	4	72308
City_Category	550068	3	B	231173
Stay_In_Current_City_Years	550068	5	1	193821
Marital_Status	550068	2	0	324731
Product_Category	550068	20	5	150933

Outlier detection

```
plt.figure(figsize=(10, 6))

# Create a box plot for 'Purchase'
sns.boxplot(x='Purchase', data=df, color='#4ce600', width=0.5)
plt.title('Boxplot for Purchase', fontsize=16)
plt.tight_layout()
plt.show()
```




```
# Calculate quartiles and IQR for the specified column
Q1 = np.percentile(df['Purchase'], 25)
Q3 = np.percentile(df['Purchase'], 75)
IQR = Q3 - Q1

# Upper and lower bounds for outliers
upper_bound = Q3 + (1.5 * IQR)
lower_bound = Q1 - (1.5 * IQR)

# Outliers in the specified column
upper_outliers_df = df[df['Purchase'] > upper_bound]
lower_outliers_df = df[df['Purchase'] < lower_bound]
```

```
# Count of outliers
upper_count = len(upper_outliers_df)
lower_count = len(lower_outliers_df)

total_count = upper_count + lower_count
```

```
print(f"Upper Outliers Count: {upper_count}")
print(f"Lower Outliers Count: {lower_count}")
print(f"Overall Outliers Count: {total_count}")
```

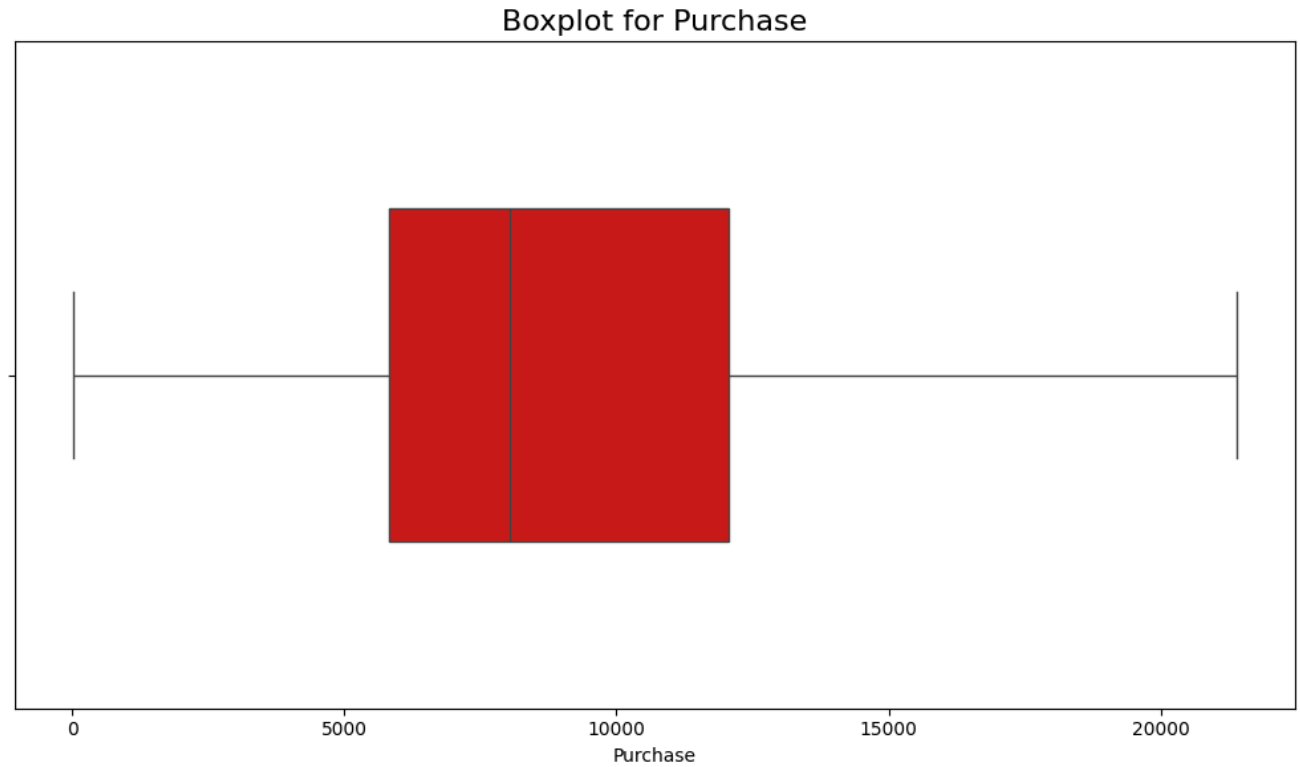
```
Upper Outliers Count: 2677
Lower Outliers Count: 0
Overall Outliers Count: 2677
```

```
# Extract rows where 'Purchase' values are greater than the upper bound to identify outliers

outliers_df = df[df['Purchase'] > upper_bound]
outliers_df
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category
343	1000058	P00117642	M	26-35	2	B	3	0	1
375	1000062	P00119342	F	36-45	3	A	1	0	1
652	1000126	P00087042	M	18-25	9	B	1	0	1
736	1000139	P00159542	F	26-35	20	C	2	0	1
1041	1000175	P00052842	F	26-35	2	B	1	0	1
...	...	...	...	...	...	...	...	...	...
544488	1005815	P00116142	M	26-35	20	B	1	0	1
544704	1005847	P00085342	F	18-25	4	B	2	0	1

```
clipped_data = np.clip(df['Purchase'], lower_bound, upper_bound)
```

```
plt.figure(figsize=(10, 6))

# Create a box plot for clipped data
sns.boxplot(x=clipped_data,color='#e60000', width=0.5, orient='h')
plt.title('Boxplot for Purchase', fontsize=16)
plt.tight_layout()
plt.show()
```




```
# Map numerical values in 'Marital_Status' to categorical labels\

df['Marital_Status'] = df['Marital_Status'].apply(lambda x: 'Married' if x == 1 else 'Single')
```

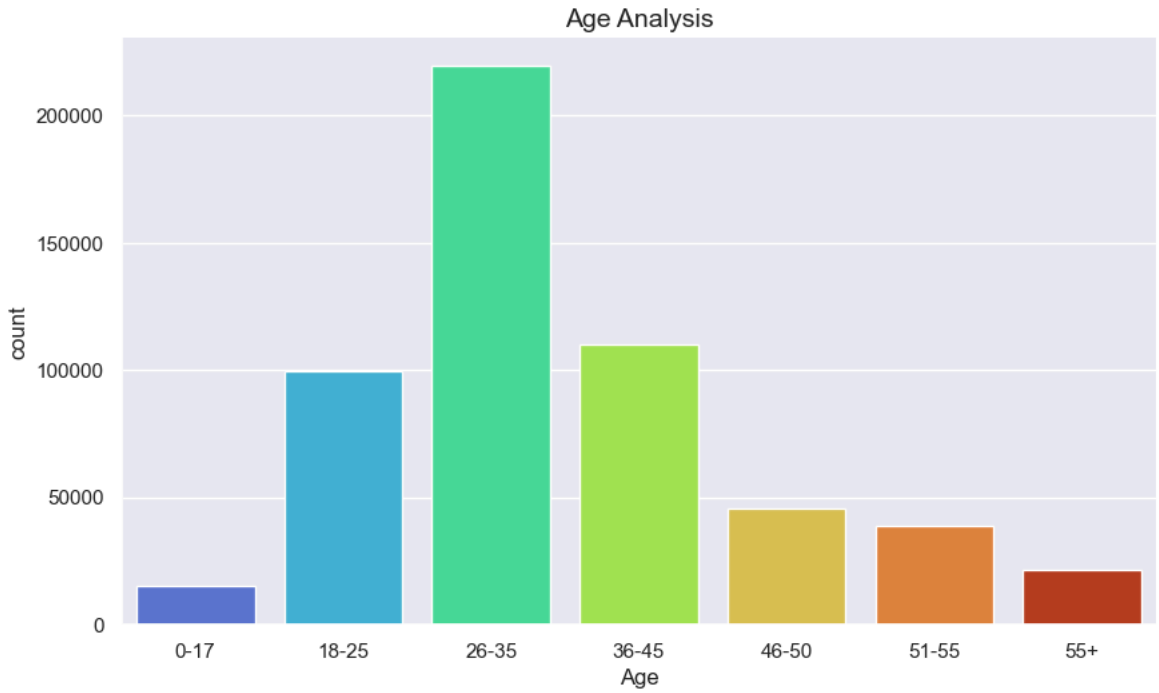
```
df.head(3)
```

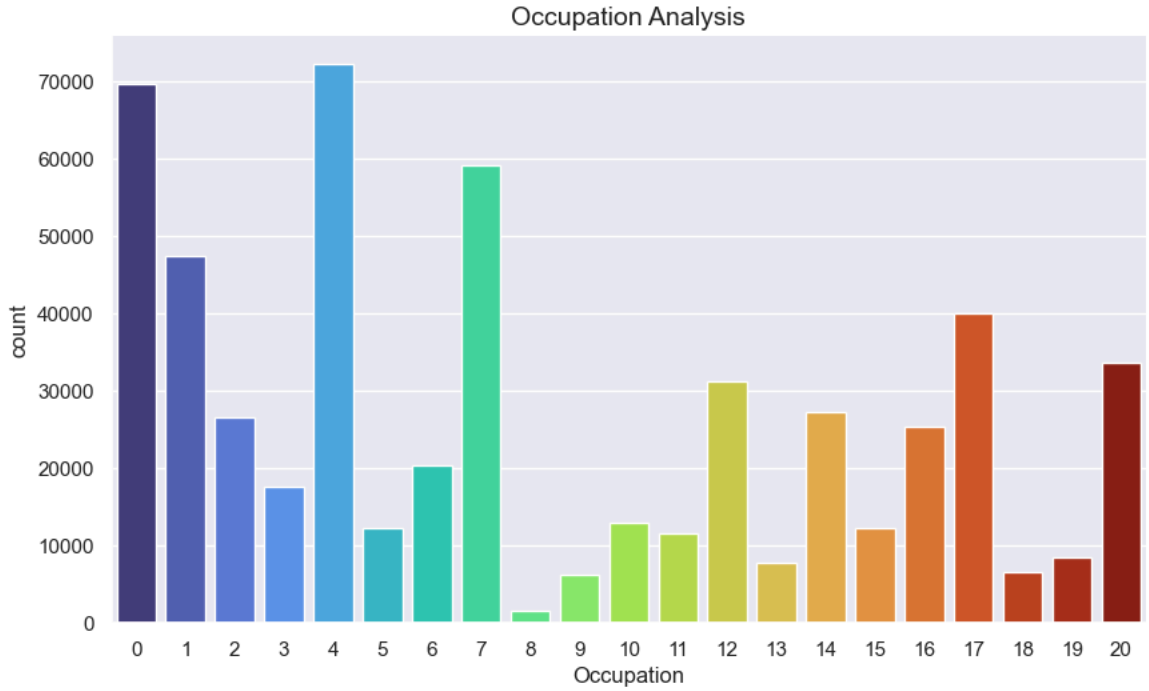
	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Pur
0	1000001	P00069042	F	0-17	10	A	2	Single	3	
1	1000001	P00248942	F	0-17	10	A	2	Single	1	

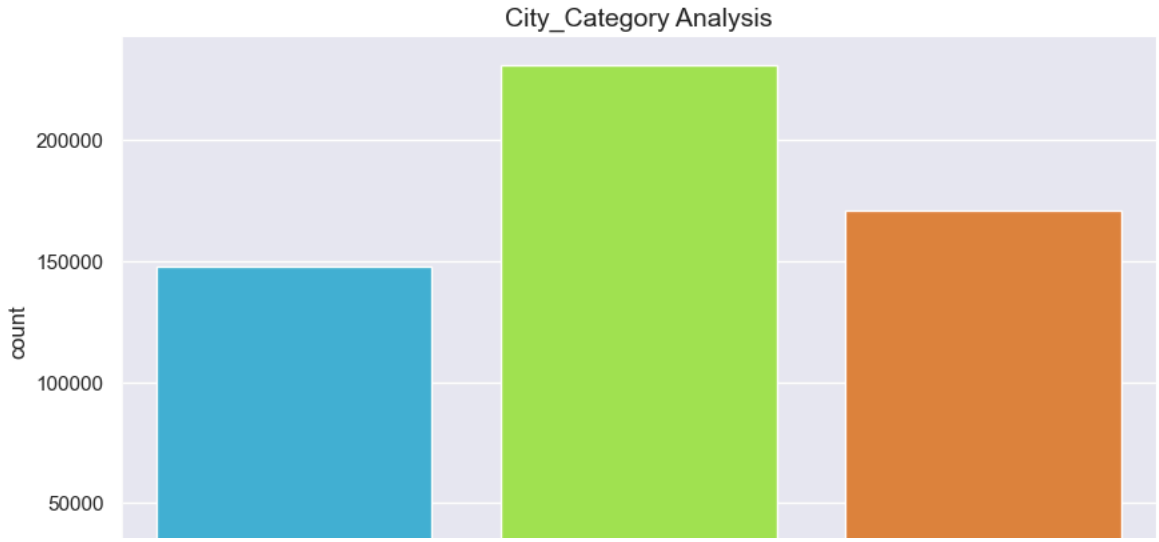
Univariate Analysis

```
category = ['Age', 'Occupation', 'City_Category', 'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category']
```

```
plt.figure(figsize=(10,40))
sns.set(style='darkgrid')

# Plot each categorical column
for i, col in enumerate(category, 1):
    plt.subplot(6, 1, i)
    sns.countplot(data=df, x=col, hue=col, palette='turbo', legend=False)
    sns.despine()
    plt.title(f'{col} Analysis', fontsize=14, fontfamily='sans-serif')

# Show the plot
plt.show()
```

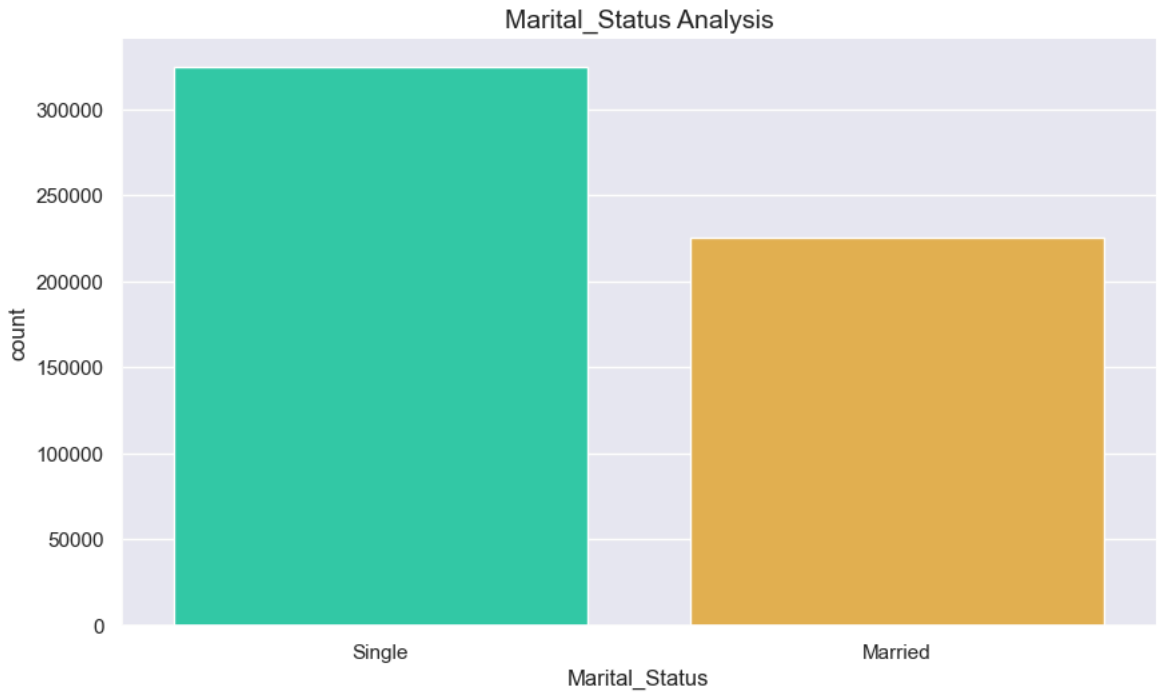

Age Analysis
count
200000
150000
100000
50000
0
0-17
18-25
26-35
36-45
46-50
51-55
55+
Age

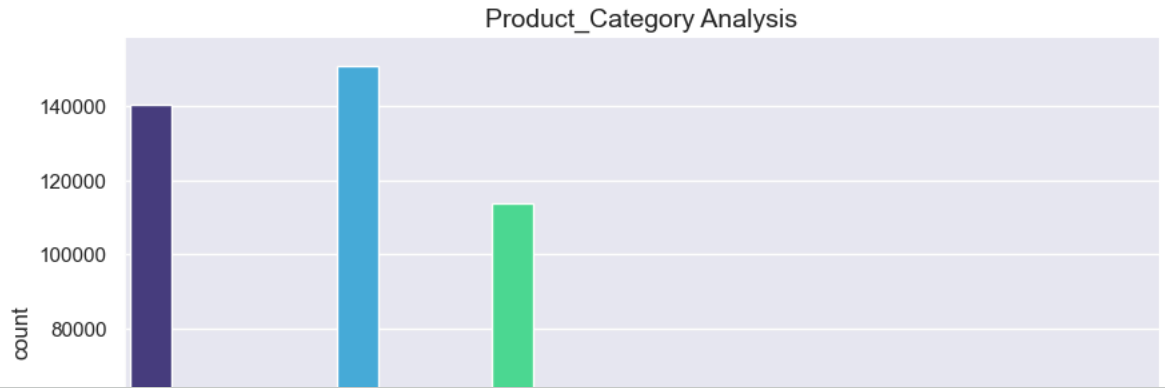

Occupation Analysis
count
70000
60000
50000
40000
30000
20000
10000
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Occupation


City_Category Analysis
count
200000
150000
100000
50000

0
A
B
C
City_Category

Stay_In_Current_City_Years Analysis
200000
175000
150000
125000
100000
75000
50000
25000
0
count
0
1
2
3
4+
Stay_In_Current_City_Years


Marital_Status Analysis
300000
250000
200000
150000
100000
50000
0
count
Single
Married
Marital_Status


Product_Category Analysis
140000
120000
100000
80000
count

Insights:

Age Group Distribution:

- The age group '26-35' has the highest count, indicating that customers in this age range make the most purchases. It is followed by the age groups '36-45' and '18-25'.

Occupation Analysis:

- Occupation '4' has the highest count, suggesting that customers with occupation '4' have the highest representation in the dataset. Occupations '0', '7', and '1' also have significant counts.

City Category Distribution:

- City_Category 'B' has the highest count, indicating that customers from City_Category 'B' have made the most purchases. City_Category 'C' and 'A' follow in terms of count.

Marital Status Impact:

- Customers with a marital status of 'Single' have a higher count compared to those who are 'Married', suggesting that single individuals make more purchases in the dataset.

City Residence Duration Impact:

- Customers who have stayed in their current city for more than 1 year show a higher purchase tendency, suggesting a positive correlation between the duration of stay and purchasing behavior.

Product Category Purchase Analysis:

- Product categories '1' and '5' exhibit higher purchase amounts, indicating that these categories contribute significantly to the overall sales revenue.

Bivariate Analysis

```
pivot = lambda index: df.pivot_table(index=df[index], columns='Gender', aggfunc='size', fill_value=0)
```

```
pivot('Age')
```

Gender	F	M
Age		
0-17	5083	10019
18-25	24628	75032
26-35	50752	168835
36-45	27170	82843
46-50	13199	32502
51-55	9894	28607
55+	5083	16421

```
pivot('Occupation')
```

Gender	F	M
Occupation		
0	18112	51526
1	17984	29442
2	8629	17959
3	7919	9731
4	17836	54472
5	2220	9957
6	8160	12195
7	10028	49105
8	361	1185
9	5843	448
10	4003	8927
11	1500	10086
12	3469	27710
13	1498	6230
14	6763	20546
15	2390	9775
16	4107	21264
17	3929	36114
18	230	6392
19	2017	6444
20	8811	24751

```
pivot('City_Category')
```

Gender	F	M
City_Category		
A	35704	112016
B	57796	173377
C	42309	128866

```
pivot('Stay_In_Current_City_Years')
```

Gender	F	M
Stay_In_Current_City_Years		
0	17063	57335
1	51298	142523
2	24332	77506
3	24520	70765
4+	18596	66130

```
pivot('Marital_Status')
```

Gender	F	M
Marital_Status		
Single	78821	245910
Married	56988	168349

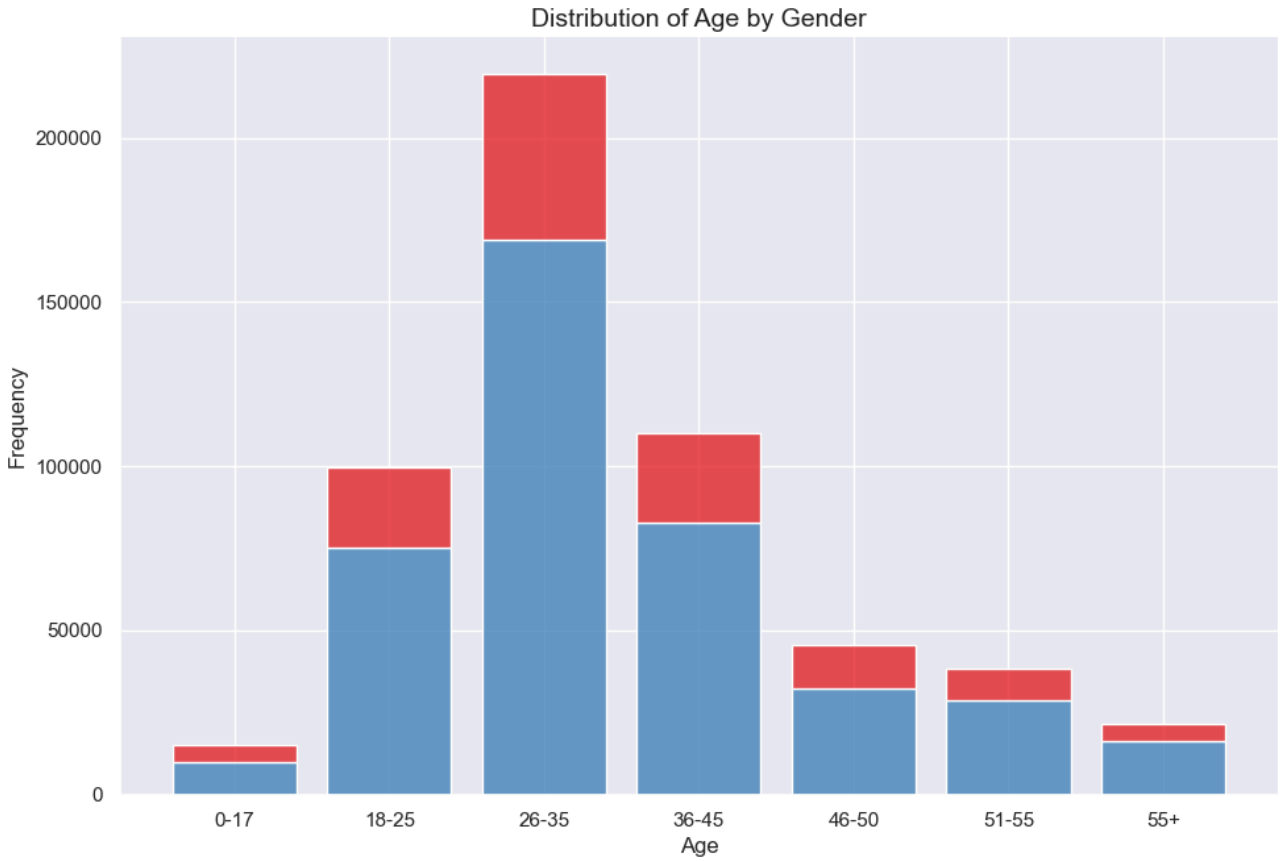
```
df.columns
```

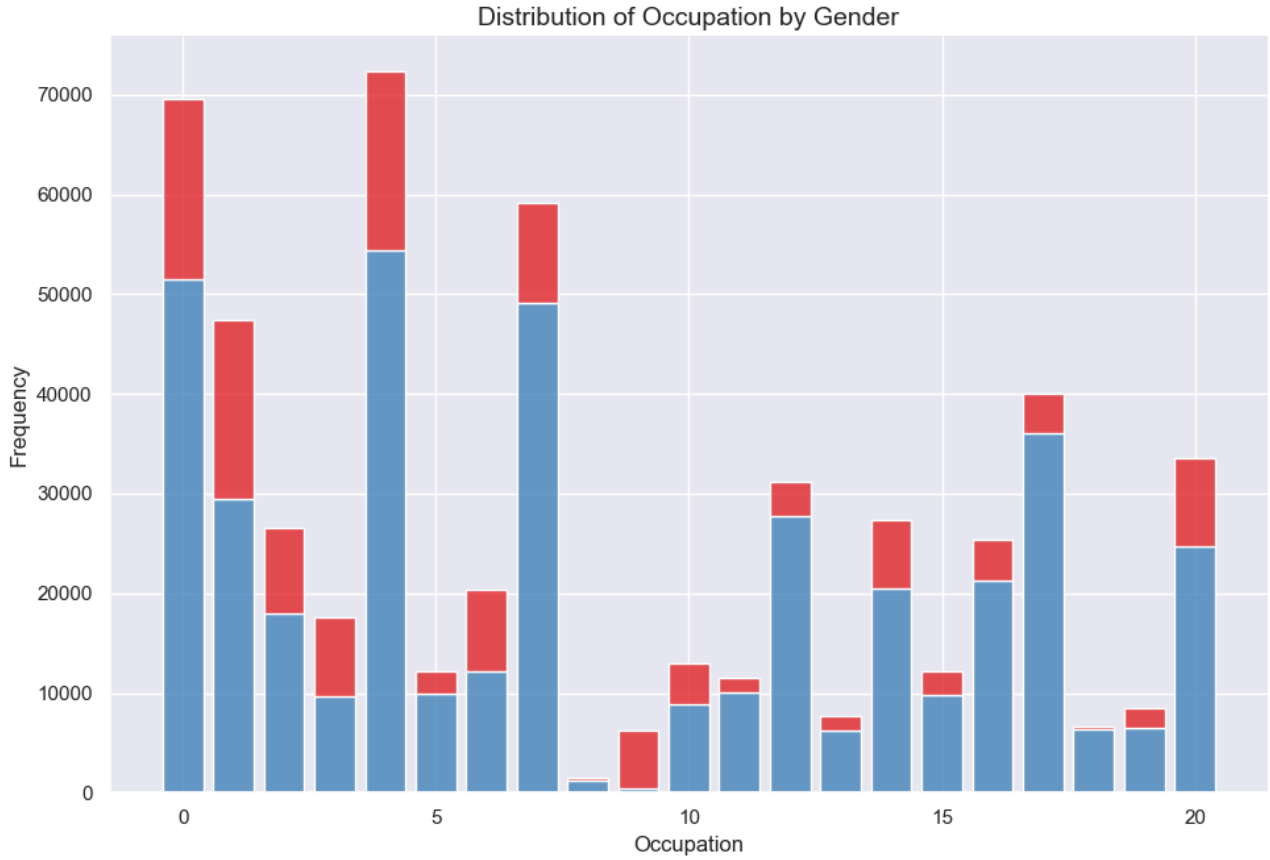
```
Index(['User_ID', 'Product_ID', 'Gender', 'Age', 'Occupation', 'City_Category',
       'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category',
       'Purchase'],
      dtype='object')
```

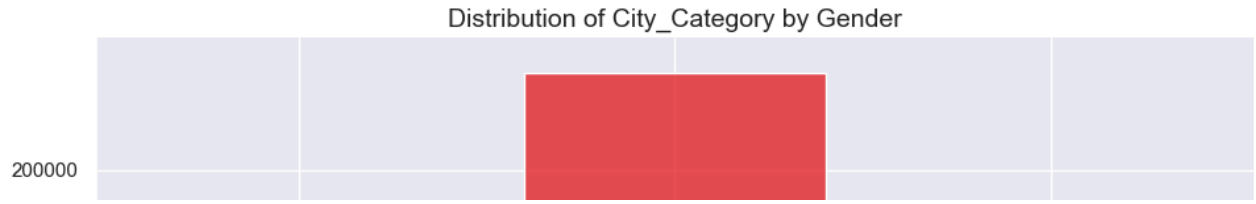
```
pivot('Product_Category')
```

Gender	F	M
Product_Category		
1	24831	115547
2	5658	18206
3	6006	14207
4	3639	8114
5	41961	108972
6	4559	15907
7	943	2778
8	33558	80367
9	70	340
10	1162	3963
11	4739	19548
12	1532	2415
13	1462	4087
14	623	900
15	1046	5244
16	2402	7426
17	62	516
18	382	2743
19	451	1152
20	723	1827

```
plt.figure(figsize=(10, 40))
sns.set(style='darkgrid')

# Plot each categorical column
for i, col in enumerate(category, 1):
    plt.subplot(6, 1, i)
    sns.histplot(data=df, x=col, hue='Gender', palette='Set1', legend=False, multiple='stack', shrink=0.8)
    sns.despine()

    # Set labels and title
    plt.xlabel(f'{col}', fontsize=12)
    plt.ylabel('Frequency', fontsize=12)
    plt.title(f'Distribution of {col} by Gender', fontsize=14, fontfamily='sans-serif')

    plt.tight_layout()

plt.show()
```

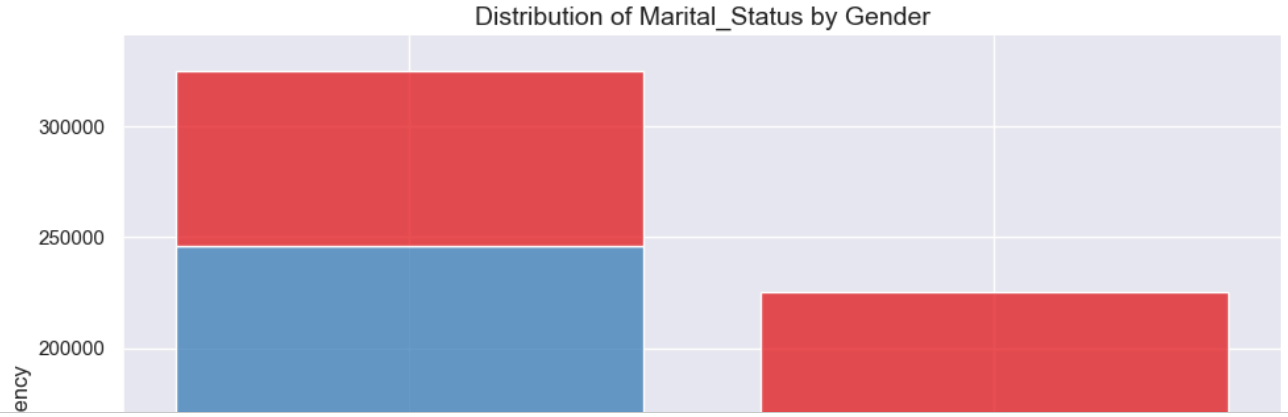

Distribution of Age by Gender
Frequency
200000
150000
100000
50000
0
0-17
18-25
26-35
36-45
46-50
51-55
55+
Age


Distribution of Occupation by Gender
Frequency
70000
60000
50000
40000
30000
20000
10000
0
0
5
10
15
20
Occupation


Distribution of City_Category by Gender
200000

150000
100000
50000
0
Frequency
A
B
C
City_Category

Distribution of Stay_In_Current_City_Years by Gender
200000
175000
150000
125000
100000
75000
50000
25000
0
Frequency
0
1
2
3
4+
Stay_In_Current_City_Years


Distribution of Marital_Status by Gender
300000
250000
200000

Insights:

Gender-Related Purchase Analysis:

- Across various age groups, males tend to have higher purchase counts compared to females, with the age group '26-35' showing the most significant difference.

Occupation-Related Purchase Analysis:

- Occupations '0' and '4' show the highest purchase counts, suggesting that individuals in these occupations contribute significantly to overall sales, with '4' having notably higher purchases than others.

City Category-Related Purchase Analysis:

- City_Category 'B' has the highest purchase counts for both genders, indicating that customers residing in City_Category 'B' contribute significantly to overall sales compared to 'A' and 'C'.

Stay in Current City Duration Impact:

- Customers who have stayed in their current city for 1 year exhibit the highest purchase counts, suggesting that individuals with a 1-year residence duration have a higher tendency to make purchases compared to other durations.

Marital Status-Related Purchase Analysis:

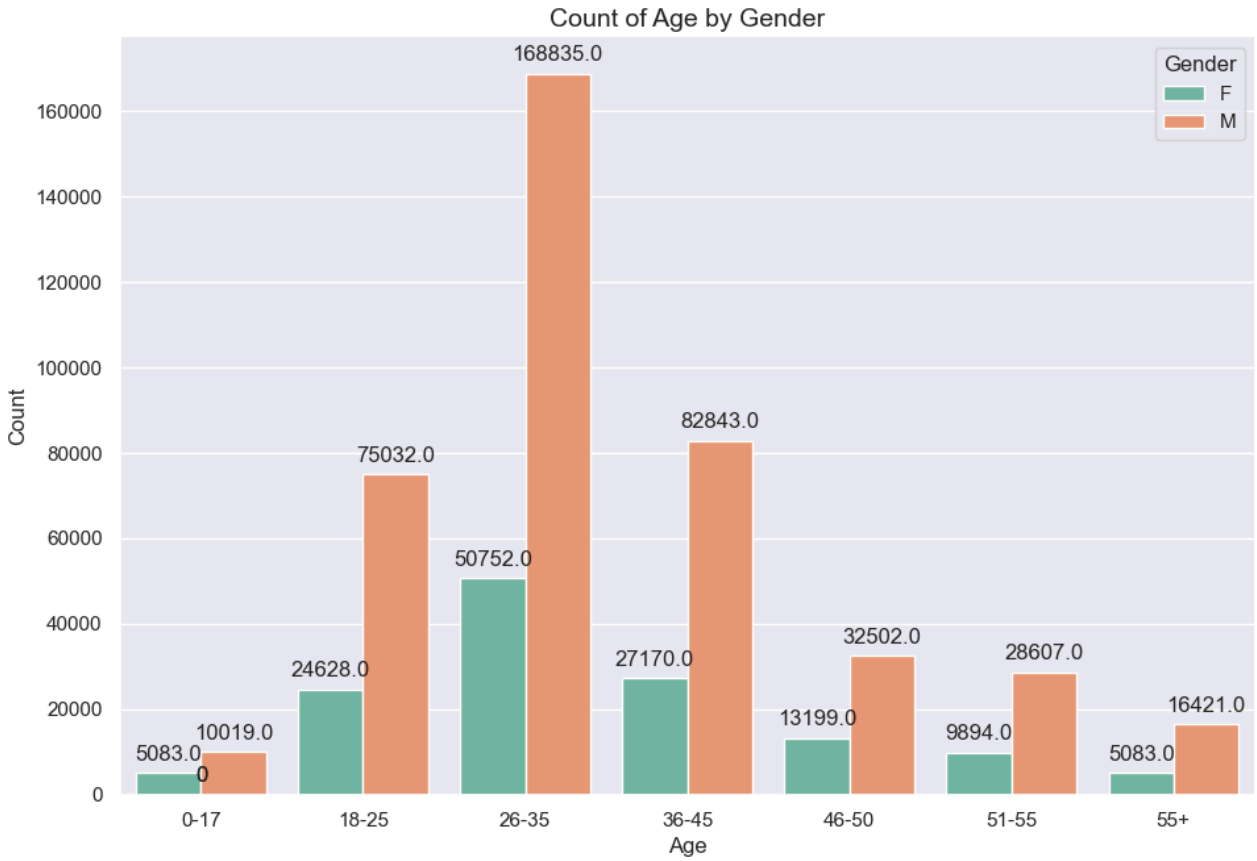
- Individuals with a marital status of 'Single' have higher purchase counts compared to those who are 'Married', indicating that single individuals contribute more to overall sales.

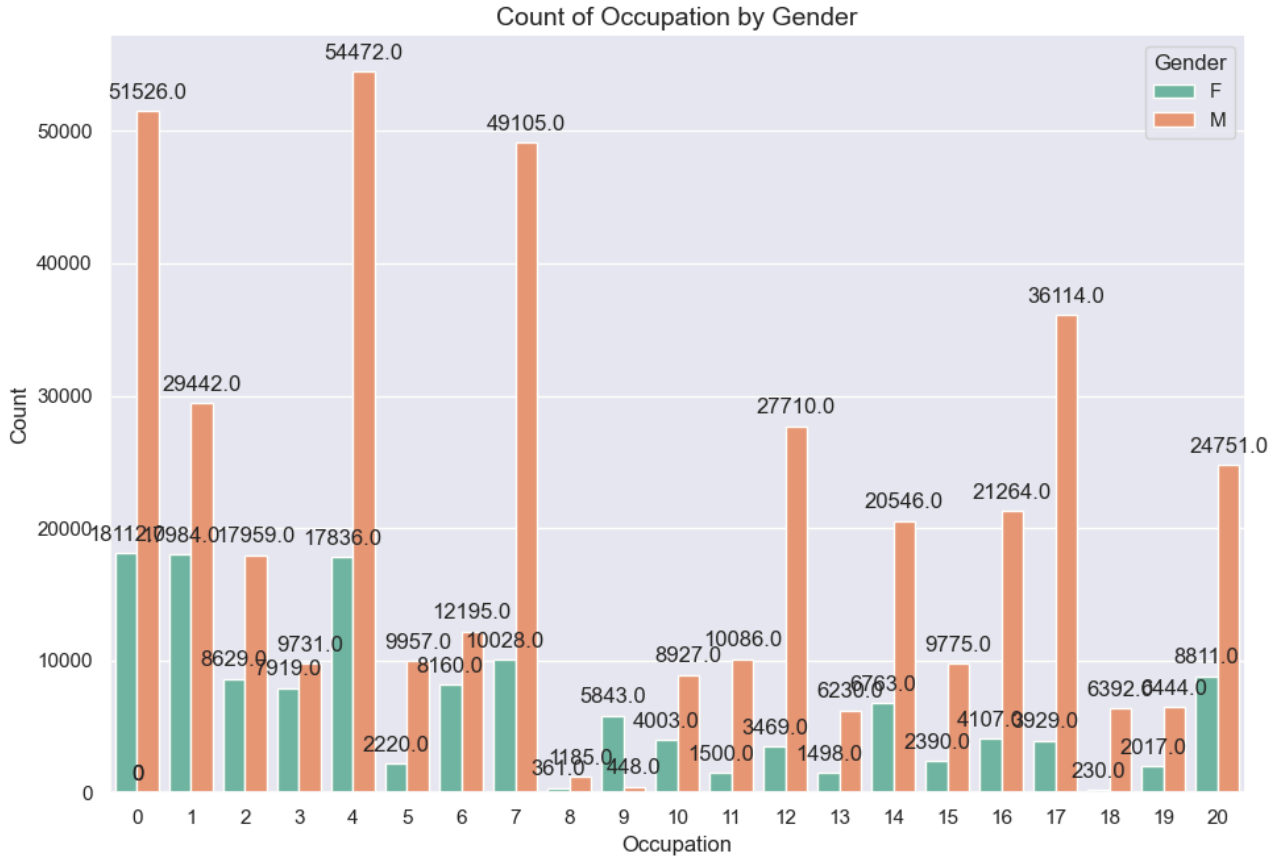
Product Category-Related Purchase Analysis:

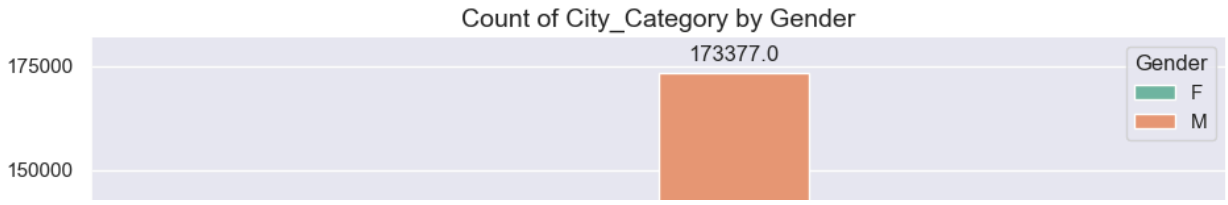
- Product Category '1' has the highest purchase counts, indicating that it significantly contributes to overall sales. Product Categories '5' and '8' also show notable purchase counts.

Multivariate Analysis

```
plt.figure(figsize=(10, 40))
sns.set(style='darkgrid')

# Plot each categorical column
for i, col in enumerate(category, 1):
    plt.subplot(6, 1, i)
    ax = sns.countplot(data=df, x=col, hue='Gender', palette='Set2')
    sns.despine()

    plt.title(f'Count of {col} by Gender', fontsize=14, fontfamily='sans-serif')
    plt.xlabel(col)
    plt.ylabel('Count')

    # Add bar counts as text labels
    for p in ax.patches:
        ax.annotate(f'{p.get_height()}',
                    (p.get_x() + p.get_width() / 2.,
                     p.get_height()),
                    ha='center',
                    va='center',
                    xytext=(0, 10),
                    textcoords='offset points')

    plt.tight_layout()

plt.show()
```


Count of Age by Gender
Gender
F
M
Count
Age
0-17
18-25
26-35
36-45
46-50
51-55
55+
5083.0
0
10019.0
24628.0
75032.0
50752.0
168835.0
27170.0
82843.0
13199.0
32502.0
9894.0
28607.0
5083.0
16421.0


Count of Occupation by Gender
Gender
F
M
Count
Occupation
51526.0
29442.0
54472.0
49105.0
27710.0
36114.0
24751.0
20546.0
21264.0
17959.0
17836.0
12195.0
10028.0
9957.0
9731.0
8629.0
7919.0
8160.0
5843.0
4003.0
8927.0
10086.0
3469.0
6763.0
9775.0
8811.0
2220.0
361.0
1185.0
448.0
1500.0
1498.0
2390.0
230.0
2017.0
0


Count of City_Category by Gender
Gender
F
M
173377.0